

CISCO PACKET TRACER TASKS - LAB # 02

Design and Simulation of Basic Networks with DHCP in Cisco Packet Tracer

[Cisco Packet Tracer .pkt file link](#)

TASK # 01 — CONCEPTUAL QUESTIONS

Q1. How does a DHCP server ensure that two devices do not receive the same IP address?

It tracks all assigned IPs and their lease times in an internal binding database, matching each IP to the device's MAC address. It will not lease an IP out again until its timer expires or the client explicitly releases it. Most servers also ping the address first to confirm no one is secretly using it.

Q2. A network is designed for 50 devices. What IP addressing scheme would you suggest, and why?

192.168.1.0/26 with a subnet mask of 255.255.255.192.

A '/26' provides 62 usable IP addresses ($2^6 - 2$), which fits the 50 devices comfortably with room for minor growth while avoiding the massive address waste of a full '/24'.

Q3. If two devices on the same network cannot communicate, what steps would you take to troubleshoot the problem?

- Check the physical link lights and cables.
- Run ipconfig or ip a to confirm both devices share the same subnet and don't have an IP clash.
- Ping 127.0.0.1 on each to verify the local network card/stack is functioning.
- Check if a local software firewall (like Windows Defender) is blocking incoming ICMP/ping traffic.

Q4. Suppose a DHCP server assigns an IP address that conflicts with a statically assigned IP address on another device. How can this issue be resolved?

Configure an exclusion range on the DHCP server so it skips that specific static address when handing out leases. Alternatively, switch that device to DHCP and set up a MAC address reservation so the server reserves that exact IP for it cleanly.

Q5. In a network of 20 computers connected via a hub, how would performance differ if the hub were replaced with a switch?

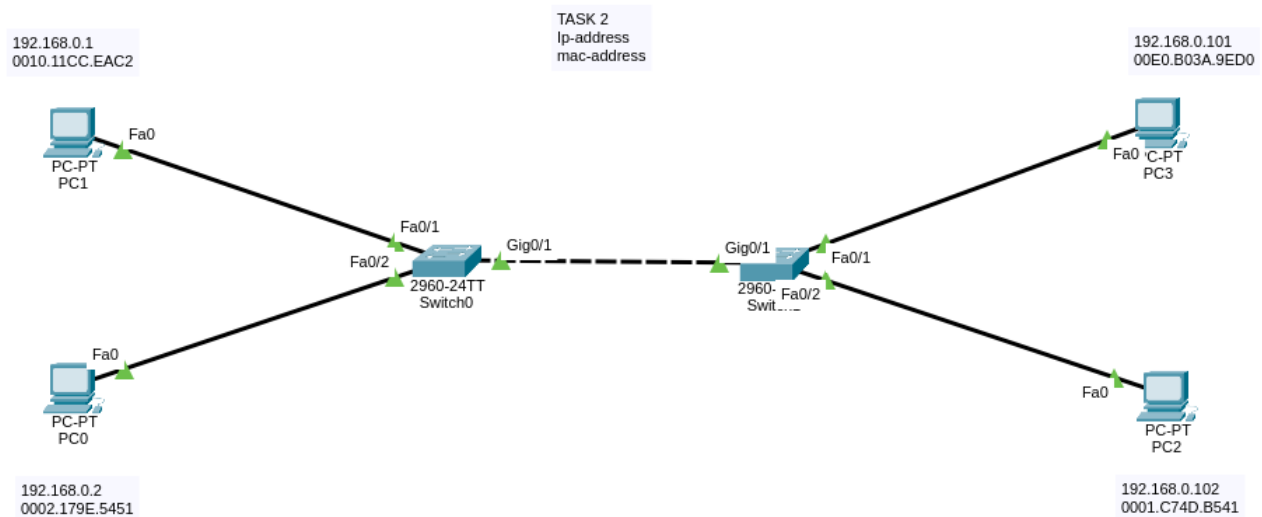
Performance will jump significantly. A hub shares one collision domain in half-duplex and broadcasts every frame to all 20 ports. A switch isolates each port into its own dedicated collision domain, orwards unicast frames only to the target device and enables full-duplex transmission without frame collisions.

Q6. Explain how DHCP works when a computer is switched on and connected to the network for the first time.

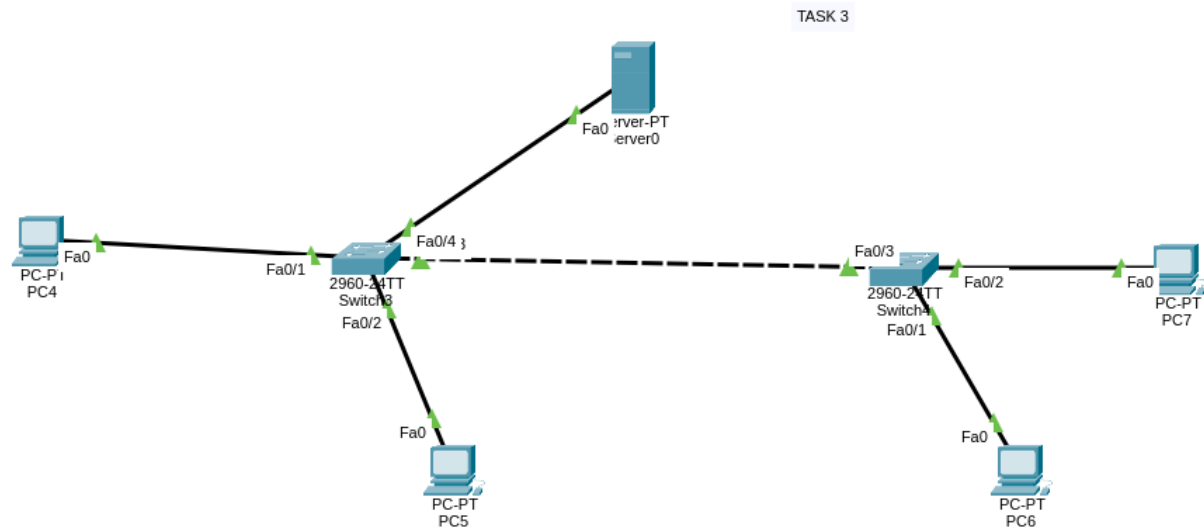
It follows the standard 4-step DORA sequence:

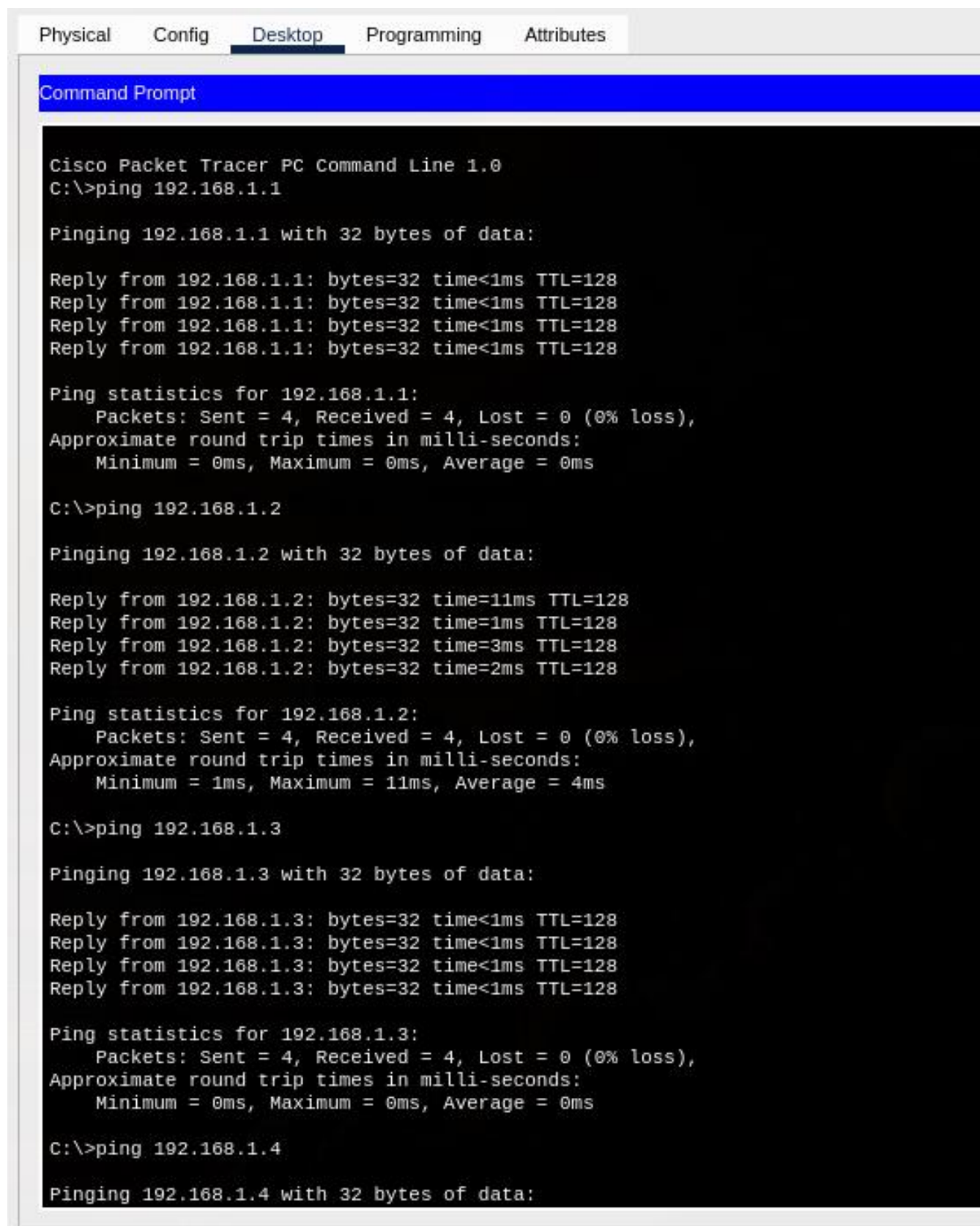
- Discover: The PC broadcasts a DHCPDISCOVER packet searching for any available DHCP server.
- Offer: The server responds with a DHCPOFFER containing an available IP, subnet mask, gateway, and DNS info.
- Request: The PC broadcasts a DHCPREQUEST asking to formally lease the offered IP.
- Acknowledge: The server sends back a DHCPACK confirming the lease and network details, completing the setup.

TASK # 02 — NETWORK DESIGN & CONFIGURATION



TASK # 03 — CONNECTIVITY VERIFICATION





The screenshot shows the Cisco Packet Tracer interface with the 'Desktop' tab selected. A 'Command Prompt' window is open, displaying the following text:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=11ms TTL=128
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=3ms TTL=128
Reply from 192.168.1.2: bytes=32 time=2ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 11ms, Average = 4ms

C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:
```

```
C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time=23ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 23ms, Average = 5ms

C:\>
```